

COURSE STRUCTURE

CourseCode	BCA30030				
CourseCategory	Program Major				
CourseTitle	Introduction to Machine Learning using Python				
Teaching Scheme	Lecture s	Tutorial s	Laboratory / Practical	Projec t	Total
Weekly load hours	3	-	2	-	5
Credits	3	-	1	-	4
Assessment Schema Code	TL3				

Pre-requisites:

Course Objectives:

10. To develop a comprehensive understanding of the principles machine learning
11. To develop the skills to apply essential data preprocessing and EDA techniques
12. To cultivate the ability to choose suitable machine learning technique and algorithm for real-world data analysis problems.
13. To introduce machine learning model performance metrics

Course Outcomes:

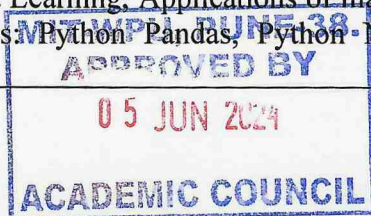
On completion of the course, student will be able to–

1. Understand the stages of machine learning project
2. Choose and apply appropriate data preprocessing and EDA techniques
3. Select and apply appropriate supervised machine learning technique and algorithm for real day problems
4. Evaluate and compare the performance of various machine learning algorithms

Course Contents:

Unit 1: Introduction to Machine Learning & Python [9]

Traditional Programming Vs Machine Learning, Steps of Machine Learning Process, Types of machine learning : Supervised Machine Learning , Unsupervised Machine Learning ,Semi-Supervised Machine Learning, and Reinforcement Learning. Applications of machine learning, Python for Data Science, Essential libraries and tools Python Pandas, Python Numpy, Python SciPy, Python Matplotlib



Unit 2: Data Preprocessing and EDA [9]

Data set : attribute types, explanatory variable, outcome variable, Data Preprocessing : Handling missing data, smooth out noise, handling outliers, encoding categorical variables, feature scaling, normalization, discretization, Why EDA? Univariate graphical EDA : Histograms, Boxplots Univariate non-graphical EDA : Frequency distribution table ,Measures, of central tendency ,measures dispersion, Multivariate non-graphical EDA: cross-tabulation, correlation matrix, Multivariate graphical EDA: Scatter plot, Heat Map, Why Feature Selection, Feature selection techniques

Unit 3: Supervised machine Learning : Regression [9]

Regression, independent variable, dependent variable, Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge Regression, Lasso Regression, Support Vector Regression, Decision Tree Regression, Random Forest Regression, Evaluating Regression models : Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared ,adjusted R- square

Unit 4: Supervised machine Learning : Classification [9]

Classification process, classification Vs Regression, Logistic Regression, Decision Tree Classification, Random Forest Classification, Support Vector Classification, Confusion matrix, Evaluating Classification models : Accuracy, Precision, Recall, F1 score

Unit 5: Unsupervised machine Learning : [9]

Clustering, Cluster analysis, Overview of clustering methods, k-means, Hierarchical clustering, Accuracy measures : Silhouette Score, Calinski-Harabasz Index, Inertia

LearningResources:

Textbooks/ReferenceBooks:

1. Introduction to Machine Learning, Ethem Alpaydin, 3rd Edition, MIT Press,2014
2. An Introduction to Machine Learning, Miroslav Kuba, 3rd Edition, Springer International Publishing,2021

Supplementary Reading:

1. Pattern Recognition and Machine Learning, C. M. Bishop, 1st Edition, Sptinger,2013
2. Introduction to Machine Learning with Python, Andreas C. Müller, Sarah Guido, 3rd Edition, O'Reilly Media,2017

Weblinks:

1. <https://www.geeksforgeeks.org/introduction-machine-learning/>
2. <https://www.javatpoint.com/machine-learning>

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc



Pedagogy:

1. Participative Learning,
2. discussions,
3. algorithm,
4. programming concepts,
5. experiential learning through practical problem solving,
6. assignments.

Laboratory Experiments / Software based Practical

Sr No.	Contents
1	Python Data Science Environment Setup, Essential libraries and tools: Python Pandas, Python Numpy, Python SciPy, Python Matplotlib
2	Implementation of EDA
3	Implementation of Supervised machine Learning : Regression
4	Implementation of Supervised machine Learning : Classification
5	Implementation of Unsupervised machine Learning

